

HYDRASAVE: A UNIQUE RIVER WATER QUALITY FORECASTING AND POLLUTION CONTROL SYSTEM

A PROJECT PHASE II REPORT
submitted by

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of

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in

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DECLARATION

We undersigned here by declare that the project report “HydraSave : A Unique River Water Quality Forecasting and Pollution Control System”, submitted for partial fulfilment of the requirements for the award of the degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of Ms. Simy Mary Kurian. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed as the basis for the award of any degree, diploma or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled **HydraSave: A Unique River Water Quality Forecasting and Pollution Control System** submitted by **Ajay Ben Abraham(AJC19CS007), Aneena Thankachan(AJC19CS022), Arya Shinod (AJC19CS038), Devarsh Damodaran (AJC19CS056)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

Technology has contributed a lot to the betterment of the environment, yet there is still a lot to be done in terms of sustaining nature and its resources. Water pollution is one of the most challenging tasks to be harnessed even with present technology. Thus, continuous monitoring and management of water resources is of prime importance. The river and its tributaries are a vital water source for millions of people. Rivers are not only an important source of water but also considered as habitat for animals and plants. Unfortunately, many rivers and streams are significantly polluted all around the world. Therefore, modelling and predicting water quality is of great importance in controlling water pollution. Water is of different quality and hence it can be used for various purposes. The water quality is measured in terms of WQI (Water Quality Index). With the calculated WQI water can be classified into five different grades. This grading allows us to know for what all purposes water of the predicted quality can be used. Even though there are many existing models for predicting water quality, there is still space for improvements as the existing models focuses only on the learning process and also do not perform missing value imputation. To overcome these problems we are proposing a model which is a combination of CNN (Convolutional Neural Network) and Bi-LSTM (Bidirectional Long Short-Term Memory) models for predicting water quality. The preprocessed data is fed into a WQI Generator whose output WQI value is used for predicting WQI for the next year. This predicted WQI is then fed into the LightGBM (Light Gradient-Boosting Machine) machine for grading the water. The dataset considered for HydraSave is from the water quality data collected from the Kerala State Pollution Control Board (KSPCB). HydraSave can be considered as a supporting hand for the pollution control campaigns conducted by the Government of India. Early prediction allows us to take the necessary steps to prevent water from getting highly polluted. Hence, our proposed model acts as a unique river water quality forecasting and pollution control system.

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ABBREVIATIONS

ANN	Artificial Neural Network
Bi-LSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
GUI	Graphical User Interface
KDE	Kernal Distribution Estimation
LightGBM	Light Gradient Boosting Machine
PCA	Principal Component Analysis
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
WQI	Water Quality Index

CHAPTER 1

INTRODUCTION

In all facets of human, environmental, and societal life, water is a fundamental necessity. The fact that water covers 71% of the world has been thoroughly examined. The rivers are significant natural water supplies. There are many uses for river water, including home and agricultural uses. Water is regarded as the most important natural resource on earth among all other natural resources due to its numerous vital uses. Hazardous wastes are, nevertheless, a result of present urbanisation and industry trends. These wastes, which might take the form of metal, organic pollutants, microbiological bacillus, and other things, are directly dumped into the rivers. Human welfare is directly impacted by water quality. The World Water Assessment Programme states that two million tonnes of human waste are disposed into water courses every day. The global rise in health problems and mortality rates, particularly in developing nations, is attributable to the widespread consumption of contaminated water. According to data from developing countries, over 250 million people contract infections annually, resulting in millions of deaths due to diseases in those regions. According to the National Water Quality Monitoring Program (NWQMP), which is run by the Central Pollution Control Board (CPCB) in collaboration with State Pollution Control Boards (SPCBs) and Pollution Control Committees (PPCs), organic pollution is the primary cause of water pollution at 2500 locations across the nation. According to the extent of organic pollution, the CPCB recognised 150 polluted river sections in 2008, and that number rose to 302 in 2015. The discharge of untreated or only partially treated sewage as well as the discharge of industrial wastewater are the

main causes of river pollution.

Hence, it is imperative to maintain regular monitoring of water quality to effectively manage its usage across various sectors. Though various models are available for water quality assessment, these need further improvement on forecasting accuracy and speed. Considering the additional requirements for better water quality forecasting we propose a model that combines CNN with Bi-LSTM to provide WQI grading prediction based on sample data followed by grading the water as per its quality using LightGBM technology. The predicted WQI value helps to analyze for what all purposes water of that quality can be used and what measures can be taken to increase the water quality.

CHAPTER 2

LITERATURE SURVEY

Various related works were analyzed from the literature being employed for water quality assessment. Conventional methods for water quality monitoring are often burdened with high implementation costs and time-consuming processes for obtaining results, necessitating further improvements in forecasting and validation. In this regard, machine learning (ML) and deep learning (DL) models have garnered attention for their ability to efficiently learn patterns and offer promising performance.

2.1 Water Quality Assessment of a River using Deep Learning Bi-LSTM Methodology: Forecasting and Validation

Authors: Sakshi Khullar and Nanhey Singh

This paper [1] propose a methodology for assessing water quality in a river using Bi-LSTM network. They trained their model using data collected from river Yamuna in India and tested its performance using various evaluation metrics. The results showed that their proposed model achieved high accuracy in predicting water quality, outperforming other traditional models. The model was tested using monthly water quality data collected at various locations in the Delhi region over a period of 6 years (2013-2019). The findings revealed that the model's predicted values were in strong concurrence with the actual

values, demonstrating its ability to accurately forecast future trends.

Advantages Deep learning models, such as Bi-LSTM, have the potential to capture complex temporal patterns and relationships in water quality data. The Bi-LSTM model captures complex dependencies and variations in water quality data, which may be challenging for traditional statistical or machine learning approaches.

Disadvantages Training and validating the model may involve long processing times and high computational costs, limiting their accessibility for researchers with limited resources. Deep learning models, including Bi-LSTM, often require large amounts of labeled training data to achieve optimal performance. Obtaining such datasets for water quality assessment can be challenging.

2.2 Groundwater Quality Assessment using Water Quality Index (WQI) in Parts of Varanasi District, Uttar Pradesh, India

Authors: Abhishek Kumar Chaurasia, H. K. Pandey and S. K. Tiwari

In drinking water quality assessment, the decision-making based on water quality data is a crucial issue because the number of parameters compromises its quality. There has been considerable advancement particularly based on the principle of WQI using slightly modified concepts. This paper [2] has presented a study where, sixteen groundwater samples were collected from the southern portion of the Varanasi district, Uttar Pradesh, India, during the pre-monsoon period of May 2015. The twenty-two water quality parameters have been considered for the calculation of water quality index namely pH, EC, total hardness (TH), total dissolved solid (TDS) and other physiochemical parameters like alkalinity, sodium, potassium calcium, lead, cadmium and many other. The BIS, 2012 has been considered to assess the suitability of groundwater for drinking purposes and for the calculation of WQI. This study showed that approximately 20% of the area is falling

under the non-suitable for drinking water category and the rest is falling under good, moderate, poor, very poor as per the WQI classification. Though ANN contains huge extra information with regard to modelled system, its performance is highly dependent on training strategy.

Advantages It is very effective for water quality monitoring. Provides means of comparisons between different rivers and sampling sites. It simplifies a complex dataset into easily understandable and usable information.

Disadvantages WQI is not an absolute measure of the degree of pollution or the actual water quality. Lack of precision and accuracy in classification technique of importance of evaluation of parameters.

2.3 Water Quality Prediction:Multi Objective Genetic Algorithm Coupled Artificial Neural Network Based Approach

Authors: :Sankhadeep Chatterjee and Sarbartha Sarkar

Water quality monitoring is an imperative process to maintain reliable and safe water's source. Water quality prediction techniques become urgent in order to test and to evaluate the water for addressing the specific water supply's problems. This ensures the accuracy of the water sources protection from prospective contamination toward achieving good quality water to preserve the overall public health .This paper [3] performs a related work in which a multi-objective genetic algorithm was employed to train the ANN to improve its performance over its traditional counterparts. The proposed model gradually minimizes two different objective functions; namely the RMSE and maximum error in order to find the optimal weight vector for the ANN. The proposed model was compared with three other, well established models namely NN-GA (ANN trained with Genetic

Algorithm), NN-PSO (ANN trained with Particle Swarm Optimization) and SVM in terms of accuracy, precision, recall, F-Measure, Matthews correlation coefficient (MCC) and Fowlkes-Mallows index (FM index). Though simulation results established superior accuracy of NN-MOGA over the other models, it still needs improvement on lowering the overall computation cost.

Advantages NN contains huge extra information with regard to the modeled system.

Disadvantages The ANN performance is highly dependent on the training strategy.

2.4 The Performance of LSTM and BiLSTM in Forecasting Time Series

Authors: :Sima Siami-Namini and Neda Tavakoli

Forecasting is an essential but challenging part of time series data analysis. The type of time series data along with the underlying context are the dominant factors affecting the performance and accuracy of time series data analysis and forecasting techniques employed. This paper paper4 reports a behavioral analysis and comparison of BiLSTM and LSTM models. The objective is to explore to what extent additional layers of training of data would be beneficial to tune the involved parameters. The results show that additional training of data and thus BiLSTM based modeling offers better predictions than regular LSTM based models. Even though it was observed that BiLSTM models provide better predictions compared to ARIMA and LSTM models, they reach equilibrium much slower than LSTM-based models.

Advantages Using BiLSTM improves the accuracy of forecast by 37.78% percent on average.

Disadvantages Training based on BiLSTM is slower and it takes fetching additional batches of data to reach the equilibrium.

2.5 Prediction of Gas Concentration Based on LSTM-LightGBM Variable Weight Combination Model

Authors: Xiangqian Wang , Ningke Xu, Xiangrui Meng and Haoqian Chang

This paper [5] proposes a model in which an improvement on accurate prediction of gas concentration was employed. This system helped to prevent gas accidents and improve the level of coal mine safety management. For considering both time series data and nonlinear characteristics of data, the system used a combination of LSTM model with the LightGBM model. The optimal parameters of gas concentration prediction were determined during the preprocessing step by the analysis of Pearson correlation coefficients of distinct sensor data. The experimental results demonstrate that the mean absolute errors of LSTM-LighGBM, LSTM and LightGBM are 1.94%, 2.19% and 2.77% respectively [4]. Thus, this study provided a novel method for gas accident prevention based on gas concentration prediction. However, it needs improvement in its performance efficiency when rapidly changing input data is fed.

Advantages Compared with that of single machine learning model and other conventional combination weighting models, the prediction result of the variable weight combination model was closer to that of the real value with a small error. It provides better prediction accuracy, and high reliability. It can give a reference for gas accidents prevention and promote the safety of coal mines.

Disadvantages The change of gas concentration is affected by complex factors and conditions. Thus it is important to consider more comprehensive factors of gas concentration, such as roof pressure, minging depth, inclination angle of coal seam and ground weather information.

2.6 A Model Combining Convolutional NeuralNetwork and LightGBM Algorithm for Ultra-short-term Wind Power Forecasting

Authors: Y. Ju, G. Sun, Q. Chen, M. Zhang, H. Zhu and M. U. Rehman

Wind energy, as the most efficient energy source, has the advantages of zero emissions, free pollution, and no fuel cost, and becomes the third largest energy source after thermal power and hydropower. However, wind power has its disadvantages, like uncertainties, which poses great challenges to power quality, grid security, and power system stability. Accurate and timely wind power forecasting has great significance for the safe operation of the grid. For the poor performance of forecasting, researchers begin to pay more attention to hybrid models, including a weather ensemble prediction framework for wind power forecast for achieving good results. This paper [6] proposed a combined model of CNN and LightGBM Algorithm for Ultra- short-term wind Power forecasting. Initially, new feature sets were constructed by analyzing the characteristics of the raw data on the time series from the wind field and adjacent wind field. Secondly, the feature extraction is done using CNN. Finally, LightGBM classification is done to improve the forecasting accuracy and robustness. The system showed better accuracy when compared with existing SVM models.

Advantages This model can solve dataset multicollinearity issues and allows fewer water quality specimens, PCR can predict the WQI more efficiently than the other techniques. The results demonstrate 95% prediction accuracy for the principal component regression method and 100% classification accuracy for the Gradient Boosting Classifier method, which show credible performance compared with the state of-art models.

Disadvantages Despite the achievements outlined in this paper, some improvements are still possible, including collection of more training samples to make the model more stable and more progress is possible on the prediction model.

2.7 Evaluation of Water Quality Using Water Quality Index (WQI) Method and GIS in Aksu River (SW-Turkey)

Authors: Sehnaz Sener, Erhan Sener and Davraz Aysen Anthropogenic influences as well as natural processes degrade surface waters and impair their use for drinking, industry, agriculture, recreation and other purposes. So a water quality monitoring program is necessary for the protection of fresh water resources. In this paper [7] a study is completed on evaluation of water quality of the Aksu River, Turkey. In this model, physical and chemical parameters of water samples taken from 21 locations were analyzed. The analysis results were compared with maximum permissible limit values recommended by World Health Organization and Turkish drinking water standards. WQI values were calculated which was then used to determine the quality. Even though classification is done precisely, there is a chance of inaccurate results due to changing hydrogeological setting.

Advantages A numerous number of water quality parameters pH, HCO₃, Cl, SO₄, Na, Ca, Mg, COD, NO₃, NO₂, Pb, Cr and Mn were used to calculate WQI values to evaluate river water quality and thus results are pretty much accurate and precise.

Disadvantages This study was conducted by assuming that physicochemical properties of water samples taken monthly do not change significantly. This assumption can affect the final results if specific conditions like flood, drought etc occurs causing unexpected change in physicochemical properties.

2.8 Water Quality Prediction and Classification Based on Principal Component Regression and Gradient Boosting Classifier Approach

Authors: Md. Saikat Islam Khan, Nazrul Islam, Jia Uddin, Sifatul Islam and Mostofa Kamal Nasir

There are no living species on earth who can survive without water. However, water is frequently polluted because the industry has been growing every year on the back of spiralling demand, and hazardous waste is discharged into the rivers and lakes by those industries. This paper [8] used water quality index (WQI), a combination of different water quality metrics that demonstrates the water quality condition of a particular region, and is applied both prediction and classification models to predict WQI and classify the water quality status. It uses a water quality prediction model utilizing the principal component regression technique. Firstly, the WQI is calculated using the weighted arithmetic index method. Secondly, the PCA is applied to the dataset, and the most dominant WQI parameters have been extracted. Thirdly, to predict the WQI, different regression algorithms are used to the PCA output. Finally, the Gradient Boosting Classifier is utilized to classify the water quality status. The proposed system is experimentally evaluated on a Gulshan Lake-related dataset. The results demonstrate 95% prediction accuracy for the principal component regression method and 100% classification accuracy for the Gradient Boosting Classifier method, which show credible performance compared with the state-of-art models. Though PCA with SVR accuracy is 95%, if the number of components reduced, then PCA with the Multiple Linear Regression model proved to be more effective.

Advantages This model can solve dataset multicollinearity issues and allows fewer water quality specimens, PCR can predict the WQI more efficiently than the other techniques. The results demonstrate 95% prediction accuracy for the principal component regression method and 100% classification accuracy for the Gradient Boosting Classifier

method, which show credible performance compared with the state of-art models.

Disadvantages Despite the achievements outlined in this paper, some improvements are still possible, including collection of more training samples to make the model more stable and more progress is possible on the prediction model.

2.9 Water Quality Index for Assessment of Drinking Groundwater Purpose Case Study: Area Surrounding Ismailia Canal, Egypt

Authors: Hend Samir Atta, Maha Abdel-Salam Omar and Ahmed Mohamed Tawfk

The groundwater quality degradation is due to rapid urban development, industrialization, and unwise water use of agricultural water, either groundwater or surface water. As groundwater quality is affected by several factors, an appropriate study of groundwater aquifers characteristics is an essential step to state a supportable utilization of groundwater resources for future development and requirements. This paper [9] proposes a WQI determining model. WQI is an approach to identify and assess the drinking groundwater quality suitability. The analyses are based on Pearson correlation to build the relationship matrix between 20 variables EC, pH, total dissolved solids (TDS), sodium (Na), potassium (K), calcium (Ca), magnesium (Mg), chloride (Cl) and others. This model helps in determining the exact extent or degree to which variables are correlated which helps in assigning weight of each parameter. However, Pearson correlation matrix cannot determine the nonlinear relationship between variables.

Advantages This model helps in determining the exact extent or degree to which variables are correlated which helps in assigning weight of each parameter.

Disadvantages Pearson correlation matrix cannot determine the nonlinear relationship between variables.

2.10 Prediction of Water Level and Water Quality Using a CNN-LSTM Combined Deep Learning Approach

Authors: Sang-Soo Baek, Jongcheol Pyo and Jong Ahn Chun

The optimization of water resources management should take into account both quantity and quality. Not only optimizing water distribution to various sectors such as domestic, agricultural, and industrial sectors but also maintaining pollution levels within permissible limits is critical for optimization. To predict surface water quality, process-based models such as the Soil and Water Assessment Tool (SWAT) and Storm Water Management Model (SWMM) have been widely used. This paper [10] proposes a combined CNN-LSTM model. It is used for finding simulated water quality including total nitrogen, total phosphorous, and total organic carbon. In this study, CNN was used to simulate the water level and LSTM used for water quality. Though combining two deep learning models increased accuracy, only three different pollutants were investigated in this study which were very less compared to existing models.

Advantages This study suggests that the combined approach of the two deep learning techniques proposed in this study has promise as a tool in accurately simulating the water level and water quality. Also, this approach can contribute to developing effective strategies for better water sustainability and management.

Disadvantages Although this model showed the acceptable model performance, only three different pollutants were investigated in this study which were very less compared to existing process-based models.

2.11 Prediction of Water Quality Index using Artificial Neural Network and Multiple Linear Regression Modelling Approach in Shivganga River basin, India.

Authors: A. K. Kadam, V. M. Wagh, A. A. Muley, B. N. Umrikar, R. N. Sankhua

In this paper [11] a study has done in which ANN and multiple linear regression (MLR) model is used to predict the water quality of Shivganga river. For predicting an accurate WQI, Levenberg–Marquardt three-layer back propagation algorithm is used. The predicted output is compared with the MLR model to check efficiency. Even though results show that the ANN predictions provide satisfactory outputs over the MLR, it has a disadvantage of overfitting and higher computation burden.

Advantages ANNs can model non-linear relationships between inputs and outputs, whereas MLR is limited to linear relationships. ANNs can handle a large number of input features, whereas MLR may struggle with high-dimensional data. ANNs are more robust to noisy data and can handle missing values better than MLR.

Disadvantages ANNs are prone to overfitting if they have too many hidden layers or neurons, while MLR is less likely to overfit. MLR models are more interpretable and easy to understand, whereas ANNs can be difficult to interpret and understand.

2.12 A Review of Water Quality Index Models and their Use for Assessing Surface Water Quality.

Authors: Md. Galal Uddin, Stephen Nash, Agnieszka I. Olbert

This paper [12] presents a comparative study of different WQI models including the differ-

ent model structures, components, and applications. Focus is placed on parameterization of the models, the techniques used to determine the sub-indices, parameter weighting values, index aggregation functions and the sources of uncertainty. The water quality index (WQI) model is a popular tool for evaluating surface water quality. It uses aggregation techniques that allow conversion of extensive water quality data into a single value or index. Issues affecting each model are also discussed.

Advantages WQI models provide a quantitative assessment of water quality, which can be useful for comparison and monitoring over time. WQI models consider multiple water quality parameters, providing a comprehensive view of water quality.

Disadvantages The choice of parameters included in the WQI can affect the results, and some parameters may be more important than others depending on the use of the water resource. The weighting of parameters in the WQI calculation can also impact the results, and different weighting schemes may be appropriate for different water resources. WQI models require accurate and complete data on water quality parameters, which may not always be available or reliable.

2.13 Using Machine Learning Models for Predicting the Water Quality Index in the La Buong River, Vietnam

Authors: Khoi DN, Quan NT, Linh DQ, Nhi PTT and Thuy NTD

This paper [13] evaluated the performance of twelve machine learning (ML) models, including five boosting-based algorithms (adaptive boosting, gradient boosting, histogram-based gradient boosting, light gradient boosting, and extreme gradient boosting), three decision tree-based algorithms (decision tree, extra trees, and random forest), and four artificial neural network (ANN)-based algorithms (multilayer perceptron, radial basis func-

tion, deep feed-forward neural network, and convolutional neural network), in predicting the surface water quality index (WQI) of the La Buong River in Vietnam. The results showed that all twelve ML models had good performance in predicting the WQI, but that extreme gradient boosting (XGBoost) had the highest accuracy. These findings suggest that ML models, especially XGBoost, can be used for WQI prediction with high accuracy.

Advantages Machine learning models have the potential to improve the accuracy of water quality predictions by analyzing large amounts of data and identifying complex patterns. ML models can assist decision-makers in managing water resources more effectively. By providing accurate predictions and insights, these models can aid in identifying pollution sources, optimizing resource allocation, and designing appropriate management strategies.

Disadvantages Some machine learning algorithms, such as deep learning models, can be highly complex and difficult to interpret. Developing and implementing machine learning models for water quality prediction requires significant computational resources and technical expertise. Adequate computing power, storage, and specialized software may be necessary to train and deploy these models.

From the analysis of existing methods designed for water quality forecasting, it can be concluded that there is still a need for performance enhancement both in terms of classification efficiency and time complexity.

CHAPTER 3

PROPOSED METHODOLOGY

This section provides a detailed description of the proposed methodology of HydraSave: A unique river water quality forecasting and pollution control system. The proposed architecture is shown in fig 3.1 .The system architecture is designed in such a way that it has less computation cost and more efficiency.

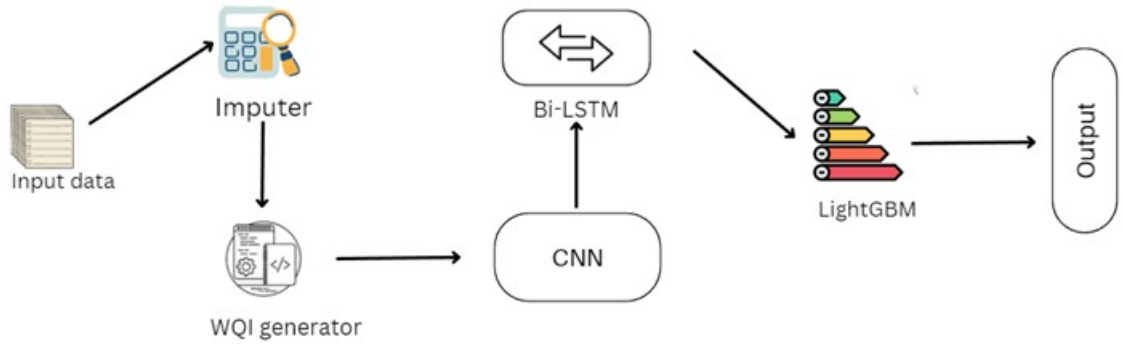


Figure 3.1: Proposed Architecture of HydraSave

Initially, the input data is preprocessed and z-score normalization is performed. This step is done to reduce errors and increase accuracy. This data is then fed into the WQI generator which produces WQI values for the complete dataset. The output values are then fed into CNN and Bi-LSTM models, where prediction of new WQI values are done. The predicted WQI values are classified using the LightGBM classifier which inturn classifies the water into any of the five different grades depending on it's quality. With the output grade achieved, the user will be able to understand for what all purposes water of

that quality can be used for the predicted year.

3.1 Input Data

The input data is collected from the Kerala State Pollution Control Board (KSPCB). The data samples of various rivers of Kerala are analyzed. The dataset used in this study includes various water quality parameters such as Dissolved Oxygen (DO), pH, Biochemical Oxygen Demand (BOD), Total Coliform (TC), and Fecal Coliform (FC), which are indicators of water quality. These parameters are used in the calculation WQI. The data thus obtained is converted into a CSV file for further preprocessing. Forecasting the quality of water is crucial as it enables the implementation of necessary pollution control measures and helps in determining the appropriate usage of water based on its quality grade. The input data collected contains water quality parameters of all stations of the river periyar considered. The time series data of the last 14 years from 2007 to 2020 is analyzed. Detailed information about input data is provided in preliminary analysis.

3.2 Data Preprocessing

Firstly, the data is passed on to an imputer where missing value imputation is done after which normalization is performed. The steps are described below.

1. Imputer : It is important to preprocess the data to avoid errors. This task is performed by the imputer. The missing value imputation technique used in this study involves replacing identified missing values with the mean of neighboring values, using a simple "mean" imputer.
2. Z-Score Normalization : A lot of machine learning algorithms attempt to find patterns in data by comparing features of data points. However, there is an issue when the features are on a completely different scales. Therefore, we are performing z-score normalization.

3.3 WQI Generator

The WQI is a significant tool used to assess the quality of drinking water in urban, rural, and industrial areas. It is calculated based on the composite influence of various water quality parameters that are considered and weighted in the calculation. The "weighted arithmetic index method" is used to compute WQI, where water quality rating (Qj) is an integral part of the index and is determined using the following expression.

$$Q_i = \Sigma q_i w_i / \Sigma w_i \text{ --- (1)}$$

Here w_i - Unit weight of i th parameter

q_i - Quality estimate scale of each parameter, it is calculated with the formula -

$$q_i = 100(V_i - V_{Ideal} / S_i - V_{Ideal}) \text{ --- (2)}$$

V_i - Measured value of i th parameter

V_{ideal} - Ideal value of i th parameter in pure water

S_i - Standard value recommended for i th parameter

w_i is calculated by the formula -

$$w_i = K / S_i \text{ --- (3)}$$

Here K is proportionality constant which is -

$$K = 1 / \Sigma S_i \text{ --- (4)}$$

Standard value recommended for parameters (S_i)-

DO, mg/L - 10

pH - 8.5

BOD, mg/L - 5

FC/100 mL - 100

TC/100 mL - 1000

Values of unit weights for each parameter (wi) -

DO -0.2213

pH - 0.2604

BOD - 0.4426

FC - 0.0221

TC - 0.0022

Ideal value of parameters (Videal)

DO -14.6

pH - 7.0

BOD - 0

FC -0

TC - 0

By using the above method, HydraSave calculates the WQI value for each input data.

3.4 Convolutional Neural Network

Convolution and pooling operations alternately make up the technique known as CNN. Each neuron executes a dot product on a few inputs, followed by an activation function. It will also have a loss or cost function on the final, completely connected layer, much like in the multilayer perceptron. CNN outperform other neural networks when given inputs such as images, voice, or audio. A standard CNN architecture used for prediction tasks typically comprises multiple layers, such as pooling layers, convolutional layers, and fully connected layers. as shown in Fig 3.2.

Convolutional layers: These layers apply convolutional filters to the input data to extract features. Each filter is a small matrix of weights that slides over the input data, computing a dot product at each position. The result is a feature map that highlights areas of the input that are important for the prediction task. Convolutional layers typically use non-linear activation functions, such as ReLU, to introduce non-linearity into the model.

Pooling layers: These layers downsample the feature maps produced by the convolutional layers. The most common pooling operation is max pooling, which takes the maximum value within a small window and discards the rest. Pooling helps feature map size reduction and make the model more computationally efficient.

Fully connected layers: These layers take the flattened feature maps from the convolutional and pooling layers and use them to make predictions. Fully connected layers are similar to the layers in a traditional neural network, with each neuron receiving input from neurons of previous layer.

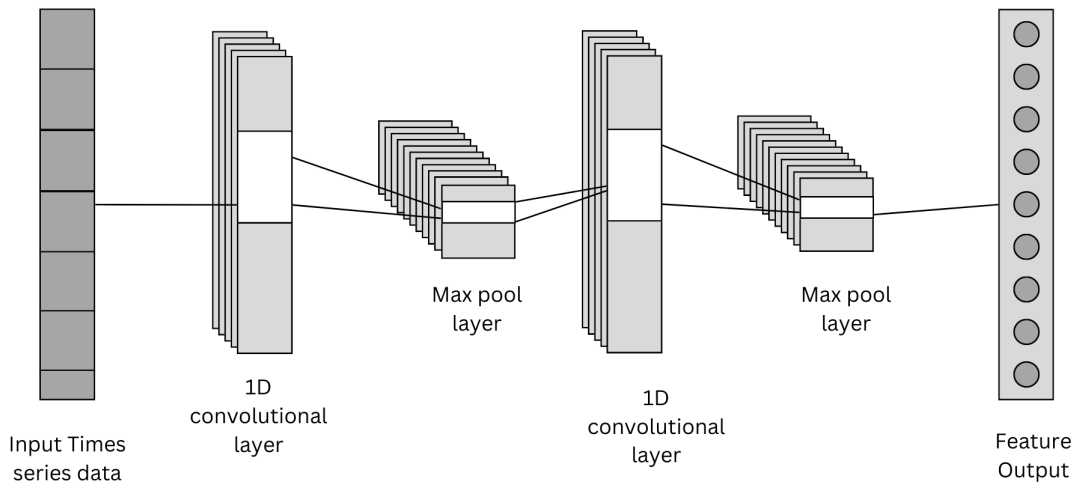


Figure 3.2: Structure of CNN

The overall architecture of a CNN for prediction tasks can vary depending on the specific task and dataset. Typically, after the convolutional and pooling layers, one or more fully connected layers are included at the end of the CNN architecture to make predictions. This architecture is often referred to as a "deep" CNN because of its many layers. Overall, CNN architecture for prediction tasks is highly effective because it allows the model to automatically learn and extract features from input data.

3.5 Bi-LSTM

Bi-LSTM is a variation of the LSTM network. Like a standard LSTM, it can process input sequences and maintain an internal state, but it can also process sequences in both directions, making it bidirectional. The input sequence is fed into the BiLSTM layer, which processes it in two directions. That is, forward and backward. The forward direction processes the input sequence from start to end, while the backward direction processes it from end to start. This allows the BiLSTM to capture context from both the past and the future. At each time step, the BiLSTM layer maintains two internal states which includes a forward state and a backward state. The forward flow is updated based on the previous forward state and the current input, while the backward flow is updated based on the previous backward state and the current input. The output of the BiLSTM layer at each time step is the concatenation of the forward and backward states as in Fig 3.3.

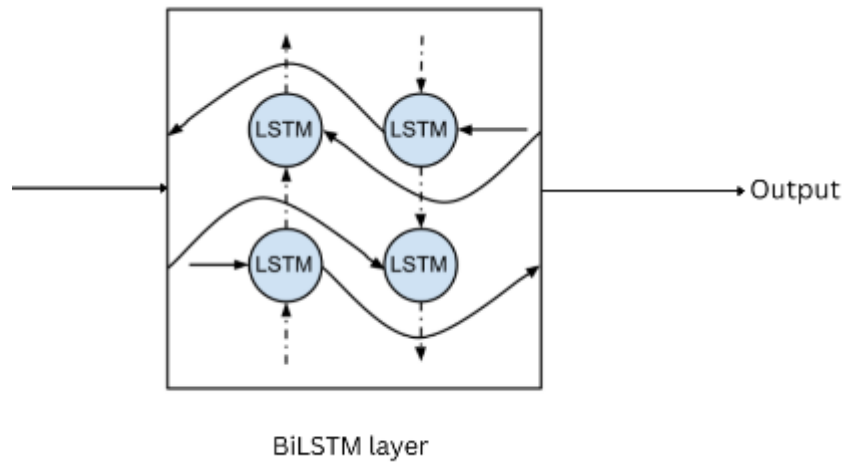


Figure 3.3: Sample Bi-LSTM Layer

The output sequence is then passed to the next layer. By incorporating bidirectional processing, the model is designed to analyze inputs in two directions - one from past to future and the other from future to past. This unique approach allows the LSTM to retain information from both the past and future, as the hidden states are combined. This enables the model to effectively capture contextual information from both directions

at any given point in time.

3.6 CNN with Bi-LSTM

In this study, a combined approach of CNN and Bi-LSTM is employed. While CNN is not well-suited for capturing long-term dependencies in time series data due to its lack of internal state, the Bi-LSTM provides a mechanism to address this limitation. CNN is utilized for extracting spatial features from the input time series data, while the Bi-LSTM is used to capture temporal dependencies. The output of the CNN serves as input to the Bi-LSTM, which generates the forecasted output as depicted in Fig 3.4. The CNN

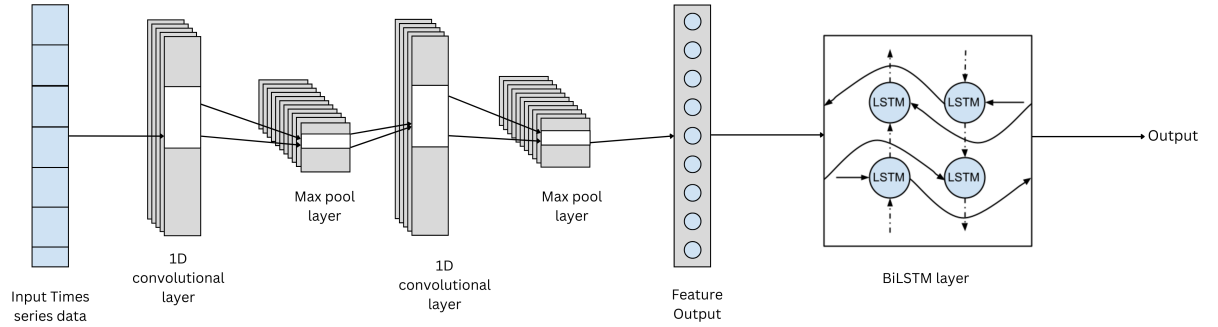


Figure 3.4: CNN and Bi-LSTM

model comprises of two one-dimensional convolutional layers followed by two max-pooling layers, which reduces computational complexity. The 1D convolutional layer generates a feature map using ReLU activation function, followed by max-pooling to downsample and select important features. The resulting output is then combined with the Bi-directional LSTM, which processes the data in both forward and backward directions, facilitating the capture of changes in input parameters.

3.7 LightGBM

LightGBM is a rapid, distributed gradient boosting framework that employs a tree-based learning algorithm. It is renowned for its support of GPU learning, making it a popular

choice for data science application development. One of the distinctive features of LightGBM is its leaf-wise splitting approach as shown in 3.5, where trees are split based on the best fit at the leaf level, in contrast to other boosting algorithms that split trees depth-wise or level-wise. This vertical tree growth strategy sets LightGBM apart, enabling faster training speed, higher efficiency, lower memory usage, improved accuracy, and support for parallel, distributed, and GPU learning. Additionally, LightGBM is capable of handling large-scale data, making it a preferred choice for various applications. XGBoost

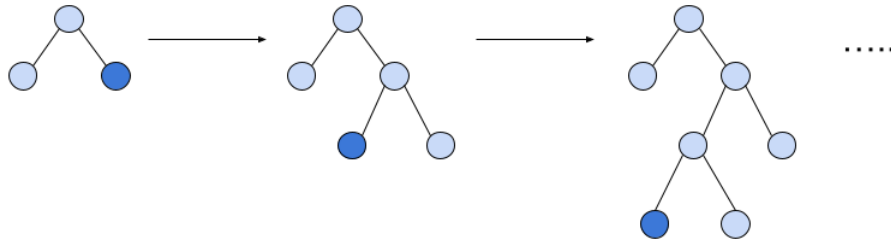


Figure 3.5: Leaf-wise Growth

utilizes a pre-sorted algorithm and a histogram-based approach for optimal split determination, whereas LightGBM employs a groundbreaking technique known as Gradient-based One-Side Sampling (GOSS) to eliminate data instances (examples or observations). This approach is faster than the pre-sorted method, which considers all potential split points on the pre-sorted feature values. However, LightGBM's GOSS technique is overall faster than the histogram-based approach.

Table 3.1: Water Quality Grading

WQI Range	Class	Water Quality
0-25	0	Excellent
25-50	1	Good
50-75	2	Medium
75-100	3	Poor
Above 100	4	Unsuitable for drinking

LightGBM classifies the predicted water quality to any of the above given grades in Table 3.1 which indicates for what all purposes water of that quality can be used.

CHAPTER 4

PRELIMINARY ANALYSIS

4.1 Dataset Collection

The concentration of fecal coliform in Periyar, as recorded by the sewage monitoring station, exceeded the allowable limit of 500mpn/100mL, with an average concentration of 113,000mpn/100mL. Thus, it is important to predict the quality of water which helps to know how much polluted the water is.



Figure 4.1: Map View of Periyar River

The sample data collected contains 5 water quality parameters namely DO, pH, BOD,

FC and TC of 8 stations for the last 6 years from 2015 to 2020. The experimental evaluation is conducted by using KSPCB dataset. Several water quality parameters collected from <https://kspcb.kerala.gov.in/water-and-air-quality-directory>.

The sampling stations are

1. Eloor
2. Kalady
3. KWA Aluva
4. SDP Aluva
5. Pathalam
6. Kalamassery
7. Purappillikkadavu
8. Muppathadam

4.2 Data Preprocessing

1. Imputer: In the dataset, SDP Aluva with station code 1338 is having missing values which is replaced by the mean of it's neighbourhood values. The values before and after imputation is shown in Table 4.1.

Table 4.1: Preprocessed Data after Imputation

Parameter	Before	After
Station	1338	1338
DO	NA	6.4
BOD	NA	1.452
pH	NA	6.622
FC	NA	24615.1
TC	NA	28360.83

2. Normalization : For normalization, we used z-score normalization to detect the outliers which are later dropped from the dataset without affecting the performance of the system.

4.3 Graphical View of The Preprocessed Output Sample

After the completion of preprocessing, the preprocessed data is used to plot KDE. It is an important tool for understanding sources of variation in the data and thereby helping to better understand the process and where root causes might be. KDE Plots depict the probability density function of the continuous or non-parametric data variable. KDE Plots are drawn for the chosen water quality parameters including BOD, DO, pH, FC and TC which are depicted below. In statistics, KDE is the application of kernel smoothing for probability density estimation, i.e., a non-parametric method to estimate the probability density function of a random variable based on kernels as weights. KDE answers a fundamental data smoothing problem where inferences about the population are made, based on a finite data sample. KDEs offer much greater flexibility because we can not only vary the bandwidth, but also use kernels of different shapes and sizes. KDE Plots are drawn for the chosen water quality parameters including BOD, DO, pH, FC and TC which are depicted below.

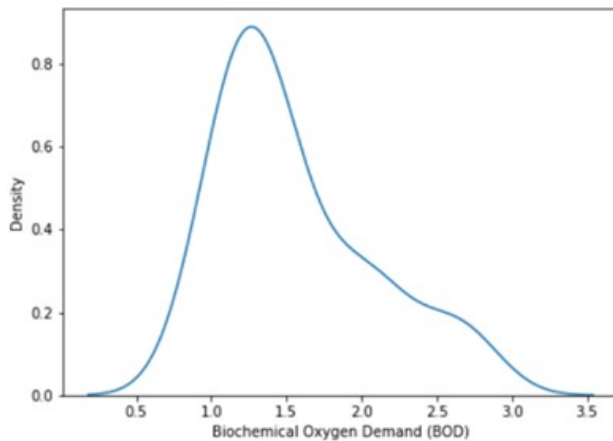


Figure 4.2: Plot for BOD

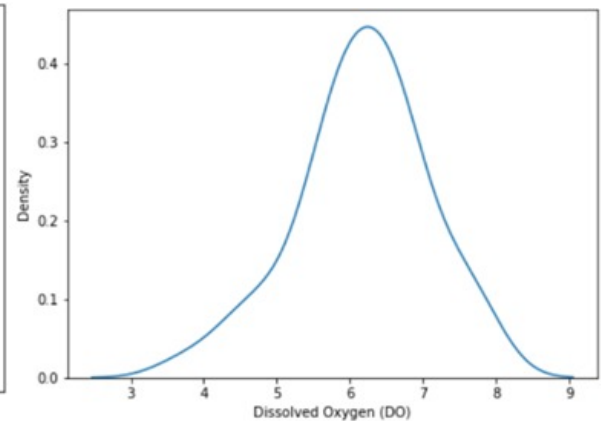


Figure 4.3: Plot for DO

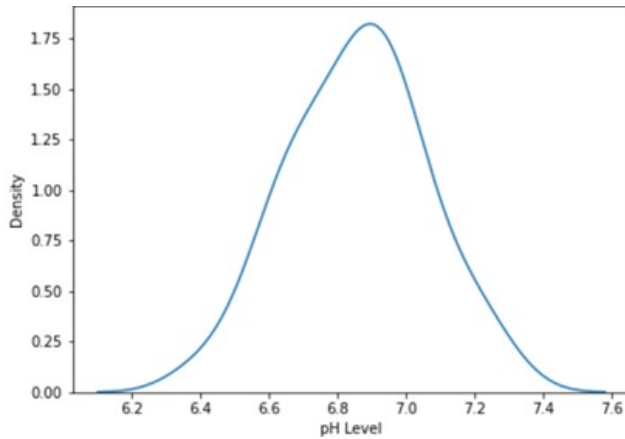


Figure 4.4: Plot for pH

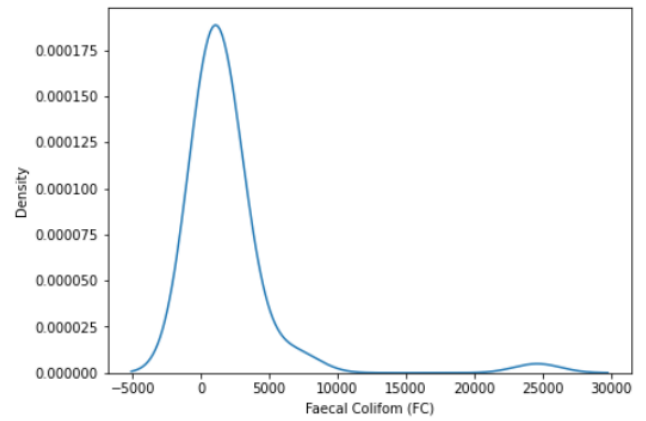


Figure 4.5: Plot for FC

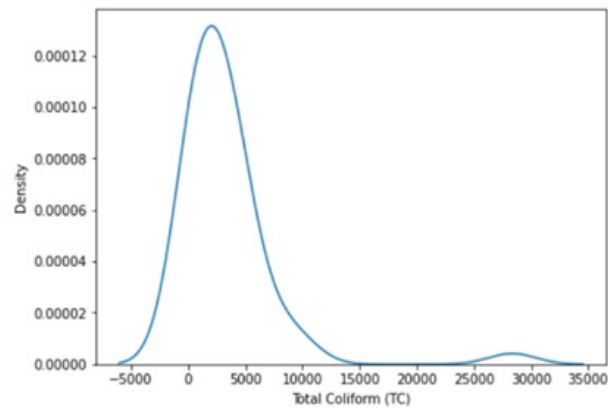


Figure 4.6: Plot for TC

4.4 WQI Generation

WQI is computed using the “weighted arithmetic index method”. A sample of the generated output is shown in Table 4.2.

Table 4.2: Sample Dataset After WQI Generation

Station	Location	WQI
18	KALADY	61.187493
3468	KWA ALUVA	47.153553
1338	SDP ALUVA	142.6759
17	ELOOR	99.822425
2333	MUPPATHADAM	57.823608
2334	PATHALAM	202.027134
2336	PURAPPILLIKADAVU	59.639521
2335	KALAMASSERY	75.552203

CHAPTER 5

GRAPHICAL USER INTERFACE

The web application of HydraSave provides a home page from which a particular river can be choosed. Each river has its own page where the functionality is available for WQI prediction.

5.1 Home Page

GUI is provided by Hydrasave WebApp. The homepage of HydraSave is provided in Fig 5.1. It shows the available rivers namely Satluj, Ganga, Godavari, Cauvery and Periyar. It also gives additional information such as different water grades and contact of the authors.

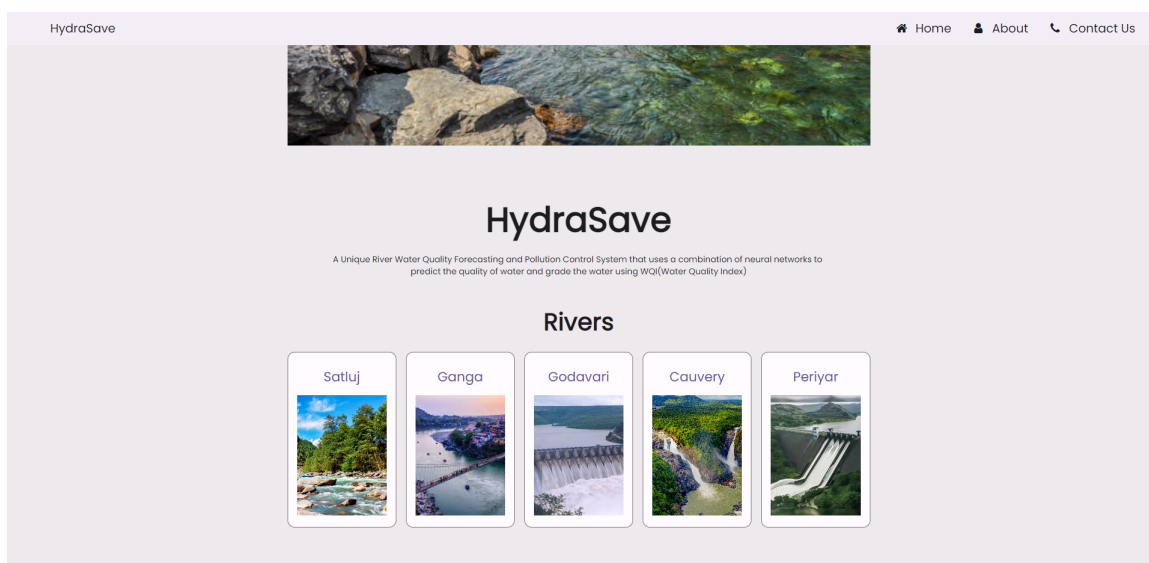


Figure 5.1: Homepage of HydraSave Webapp

5.2 Main Page of River Satluj

The Fig 5.2 shows the information of river Satluj. A drop down list is available to select stations of respective rivers. The predicted WQI value and quality grade of selected station will be displayed on the webpage. Similarly, other rivers have their own respective pages.

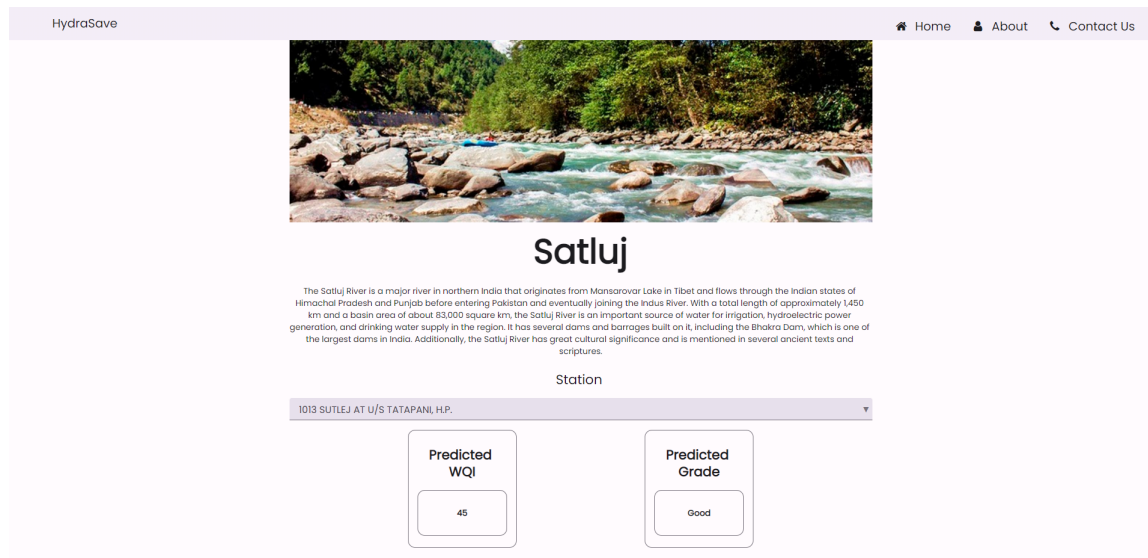


Figure 5.2: Details of River Satluj

CHAPTER 6

EXPERIMENT AND RESULT

Initially, the generated WQI values are then fed into CNN. The predicted value is again passed to Bi-LSTM for increasing prediction accuracy. The model predicts the WQI value of the year 2020 of all the 8 stations renamed as S1 to S8 respectively. The below provided graph Fig 6.1 is plotted to visualize the actual and predicted WQI value of each station respectively. The graph shows how accurate the result will be for the tested case approximately.

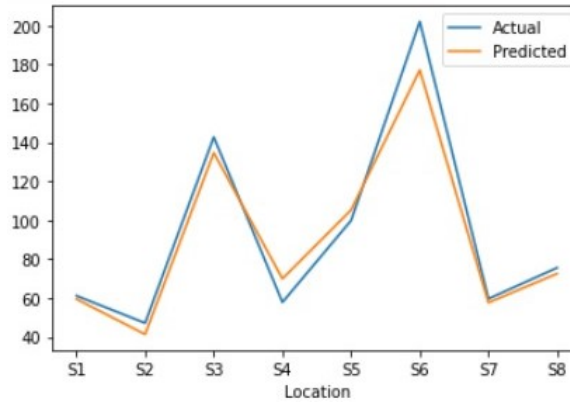


Figure 6.1: Actual and Predicted WQI Value of Each Station

With the predicted value of WQI, the LightGBM classifies the input sample into any of the five different grades. As per the predicted WQI, the LightGBM grades water quality of each station as shown in Table 6.1. With the output grade achieved, the user will be able to understand for what all purposes water of that quality is used.

The performance evaluation is done by various measures like precision, recall, F1-Score. Corresponding values are provided in Table 6.2.

Table 6.1: Sample Dataset after Quality Grading

Station	Location	Grade
18	KALADY	2
3468	KWA ALUVA	1
1338	SDP ALUVA	4
2333	MUPPATHADAM	2
17	ELOOR	4
2334	PATHALAM	4
2336	PURAPPILLIKADAVU	2
2335	KALAMASSERY	2

Table 6.2: Performance Measure of LightGBM

Class	Precision	Recall	F1-Score
0	1.00	1.00	1.00
1	1.00	1.00	1.00
2	1.00	0.91	0.95
3	0.91	1.00	0.95
4	1.00	1.00	1.00

The experimentation result showed that HydraSave requires less computation cost and is much more faster than other existing models. CNN-BiLSTM used in the system showed higher accuracy than CNN-LSTM model. LightGBM classifier has a higher classifying speed when compared to other classification models. Overall, HydraSave is capable of predicting water quality with lesser error rate and higher computation speed.

CHAPTER 7

FUTURE WORK

Though the system shows much better performance when compared, it still can upgrade its working. Some upgradation and changes are expected to be applied in the system for making it highly efficient. It includes:

1. Rather than considering river Periyar only, the input data will be extended by taking sample data of other rivers of Kerala.
2. Applying transfer learning and thus reduce the overall computation cost.
3. Trying to implement using light-weight models, so that the model can be deployed on a mobile device.

CHAPTER 8

CONCLUSION

HydraSave is a water quality forecasting model that utilizes CNN combined with Bi-LSTM networks. The model explores the applications of Bi-LSTM for forecasting WQI by incorporating data pre-processing, normalization, prediction, and classification of water quality. Traditional CNN and LSTM models often suffer from high computational complexity and low prediction accuracy. To address these limitations, a combination model is used in HydraSave, which helps reduce training errors and improves precision.

Additionally, the HydraSave model incorporates LightGBM technology for water classification based on the predicted WQI value. The model is implemented using time-series data from various monitoring stations of river Periyar in Kerala to study water quality. Accurate prediction of water quality can assist the government in implementing necessary pollution control measures, making HydraSave a unique system for water quality forecasting and pollution control.

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APPENDIX A

SAMPLE CODE FOR ML MODEL

Code for Imputer: Data preprocessing

```
df.isnull().sum().sort_values()

df_num = df.select_dtypes(exclude="object")
df_num_col = df_num.columns
imputer = SimpleImputer(strategy="mean")
df_num = imputer.fit_transform(df_num)
df_num = pd.DataFrame(df_num, columns=df_num_col)
```

Code for Z score Normalization:

```
df_num_final_norm = zscore(df_num_final, axis=0)

def indices_of_greater_than_3(df_norm):
    indices_arr = []
    n_col = df_norm.shape[1]
    for index in range(n_col):
        col_index = df_norm.iloc[:, index]
        greater_than_3 = df_norm[col_index > 3]
        greater_than_3_index = greater_than_3.index
        indices_arr.extend(greater_than_3_index)
```

```
    return indices_arr
indices_arr = indices_of_greater_than_3(df_num_final_norm)
print("Number of outliers using Z-Score method-",len(indices_arr))
df_final.iloc[indices_arr, :]
```

Code for calculating Water Quality Index of each sample:

```
df_num_final = df_final.select_dtypes(exclude="object")
df_num_final.drop(["Year"], axis=1, inplace=True)
wi = np.array([0.2213, 0.4426, 0.2604, 0.0221, 0.0022])
si = np.array([10, 5, 8.5, 100, 1000])
vIdeal = np.array([14.6, 0, 7.0, 0, 0])
def calc_wqi(sample):
    wqi_sample = 0
    num_col = 5
    for index in range(num_col):
        v_index = sample[index]
        of sample at index
        v_index_ideal = vIdeal[index]
        w_index = wi[index]
        std_index = si[index]
        q_index = (v_index - v_index_ideal) / (std_index - v_index_ideal)
        q_index = q_index * 100 # Final qi value of obeserved value
        wqi_sample += q_index*w_index
    return wqi_sample
```

Code for computing WQI for the whole dataset

```
def calc_wqi_for_df(df):
```

```
wqi_arr = []
for index in range(df.shape[0]):
    index_row = df.iloc[index, :]
    wqi_row = calc_wqi(index_row)
    wqi_arr.append(wqi_row)
return wqi_arr
wqi_arr = calc_wqi_for_df(df_num_final)
# Converting ordinary array to numpy array
wqi_arr = np.array(wqi_arr)
wqi_arr = np.reshape(wqi_arr, (-1, 1))
# Resetting index values of the dataframes
wqi_arr_df = pd.DataFrame(wqi_arr, columns=["WQI"]).reset_index()
df_final = df_final.reset_index()
# Combining dataframe of WQI and dataframe of attributes
df_wqi = pd.concat([df_final, pd.DataFrame(wqi_arr, columns=["WQI"])], axis=1)
df_wqi.drop("index", axis=1, inplace=True)
df_wqi.shape
```

Checking for negative WQI values:

```
df_wqi[(df_wqi["WQI"] < 0)]
```

Develop Machine Learning Models For Univariate Time Series Forecasting

```
import numpy as np
import pandas as pd
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM
from tensorflow.keras.layers import Dense
from tensorflow.keras.layers import Flatten
from tensorflow.keras import layers
from tensorflow.keras.layers import MaxPooling1D
from tensorflow.keras.layers import Conv1D, LSTM, Dense, Dropout, Bidirectional
```

```
#mount google drive
from google.colab import drive
drive.mount('/content/drive')

# accessing file from google drive
df=pd.read_csv("/content/drive/MyDrive/Project/input/wqioutput.csv",

encoding= 'unicode_escape')
#df = df.iloc[0:1900, :]
df.shape
df

# upload file from local system
from google.colab import files

uploaded = files.upload()

for fn in uploaded.keys():
    print('User uploaded file "{name}" with length {length} bytes'.format(
        name=fn, length=len(uploaded[fn])))

def create_sequences(timeseries_data, window_size):
    X = []
    y = []
    for i in range(len(timeseries_data) - window_size):
        X.append(timeseries_data[i:i+window_size])
        y.append(timeseries_data[i+window_size])
    return np.array(X), np.array(y)

# reshape from [samples, timesteps] into [samples, timesteps, features]
```

```
n_features = 1
n_steps= 5
X = X.reshape((X.shape[0], X.shape[1], n_features))
```

Building Machine Learning Model

```
# CNN-BiLSTM model
model = Sequential()
model.add(Conv1D(filters=64, kernel_size=3, activation='relu', input_shape=(n_steps, n_features)))
#model.add(MaxPooling1D(pool_size=2))
#model.add(Conv1D(filters=32, kernel_size=2, activation='relu',padding="same"))
#model.add(MaxPooling1D(pool_size=2))
#model.add(Flatten())
model.add(Bidirectional(LSTM(50, activation='relu', input_shape=(n_steps, n_features))))
model.add(Dense(1))
model.compile(loss='mean_squared_error', optimizer='adam') #rmsprop adam

model.summary()

# Define the CNN model
model_cnn = Sequential()
model_cnn.add(Conv1D(filters=32, kernel_size=3, activation='relu', input_shape=(n_steps, n_features)))
model_cnn.add(MaxPooling1D(pool_size=2))
model_cnn.add(Flatten())
model_cnn.add(Dense(1))

model_cnn.compile(optimizer='adam', loss='mean_squared_error')

#cnn_bilstm_gpt
model_gpt = Sequential()
model_gpt.add(Conv1D(filters=64, kernel_size=3, activation='relu',padding="same", input_shape=(n_steps, n_features)))
model_gpt.add(Dropout(0.2))
model_gpt.add(Bidirectional(LSTM(64, activation='relu'))))
```

```
model_gpt.add(Dense(1))

# Compile the model
model_gpt.compile(optimizer='adam', loss='mean_squared_error')

# Lstm model
model = Sequential()
model.add(Conv1D(filters=64, kernel_size=3, activation='relu', input_shape=(n_steps,
model.add(LSTM(50, activation='relu', return_sequences=True, input_shape=(n_steps,
model.add(LSTM(50, activation='relu'))
model.add(Dense(1))
model.compile(optimizer='adam', loss='mean_squared_error')
# fit model
model.fit(X, y, epochs=300, verbose=1)

# fit model
model.fit(X, y, epochs=400, verbose=1)

model_cnn.fit(X, y, epochs=500, batch_size=16, verbose =1)

#save model to drive
model.save('/content/drive/MyDrive/model_cnnbiltm.h5')
```

Data Prediction

```
# demonstrate prediction for next 10 years
x_input = np.array([49.50664203, 46.39254754, 49.43413101, 51.02788203, 38.02473565
temp_input=list(x_input)
lst_output=[]
i=0
while(i<10):
```

```

if(len(temp_input)>5):
    x_input=np.array(temp_input[1:])
    print("{} year input {}".format(i,x_input))
    #print(x_input)
    x_input = x_input.reshape((1, n_steps, n_features))
    #print(x_input)
    yhat = model.predict(x_input, verbose=0)
    print("{} year output {}".format(i,yhat))
    temp_input.append(yhat[0][0])
    temp_input=temp_input[1:]
    #print(temp_input)
    lst_output.append(yhat[0][0])
    i=i+1
else:
    x_input = x_input.reshape((1, n_steps, n_features))
    yhat = model.predict(x_input, verbose=0)
    print(yhat[0])
    temp_input.append(yhat[0][0])
    lst_output.append(yhat[0][0])
    i=i+1

print(lst_output)

from keras.models import load_model

model = load_model("/content/drive/MyDrive/model_cnnbilstm.h5")

d= timeseries_data[(len(timeseries_data)-n_steps):]
dx = np.array(d)
temp_input=list(dx)

```

```
dx = dx.reshape((1, n_steps, n_features))
yhat = model.predict(dx, verbose=0)
print("Predicted WQI value = {}".format(yhat))

WQIv=list(yhat[0])
#print(WQIv[0])
dict = {'WQI':WQIv}
WQIout = pd.DataFrame(dict)
#print(WQIout)
wqi_data = WQIout[['WQI']]
wqi_data
```

Code for LightGBM Classifier

```
import io
import pandas as pd
import lightgbm as lgb
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
from sklearn.model_selection import train_test_split
from pandas import DataFrame
from numpy import argmax

df = pd.read_csv("test - Sheet1.csv",header=0)
df.head()
df.dtypes
df ['STATION']=df.STATION.astype(object)
df['WQICLS'].value_counts()
x =df[['WQI ']]
y =df['WQICLS']
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.38)
params = {                                #hyperparameters
```

```
'boosting': 'gbdt',          #baselearner
'objective': 'multiclass',
'num_leaves': 10,
'num_class': 5
}
lgb_train = lgb.Dataset(x_train, y_train)
lgb_eval = lgb.Dataset(x_test, y_test, reference=lgb_train)
model = lgb.train(params,
                  train_set=lgb_train,
                  valid_sets=lgb_eval,
                  early_stopping_rounds=30)
y_pred = model.predict(x_test)

y_pred = argmax(y_pred, axis=1)
cr = classification_report(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)

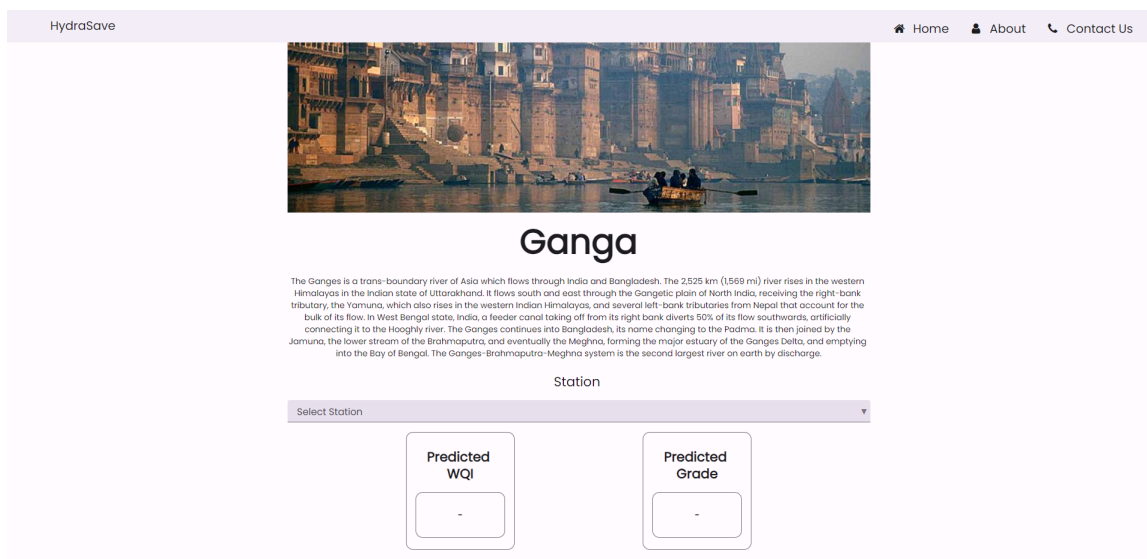
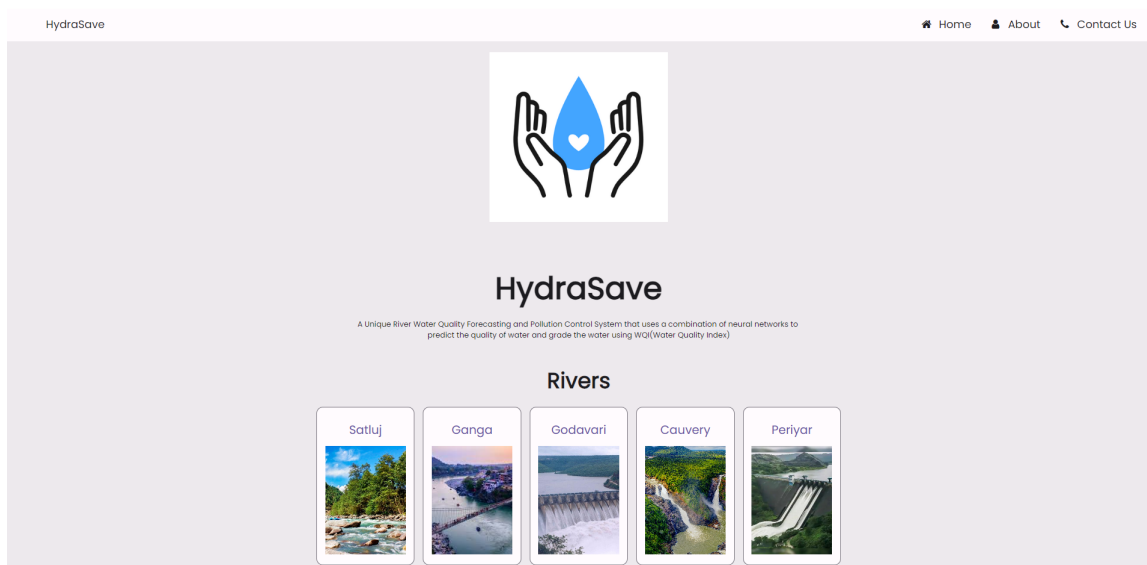
print(cr)
print(cm)

lgb.plot_importance(model, height=.5)

#saving model
model.save_model('lgbmmodel.model')
```

APPENDIX B

SCREENSHOTS



HydraSave

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Godavari

The Godavari is India's second longest river after the Ganga river and drains into the third largest basin in India, covering about 10% of India's total geographical area. Its source is in Trimbakeshwar, Nashik, Maharashtra. It flows east for 1,465 kilometres (910 mi), draining the states of Maharashtra (48.6%), Telangana (18.8%), Andhra Pradesh (4.5%), Chhattisgarh (10.9%) and Odisha (5.7%). The river ultimately empties into the Bay of Bengal through an extensive network of distributaries. Measuring up to 312,812 km² (120,777 sq mi), it forms one of the largest river basins in the Indian subcontinent, with only the Ganga and Indus rivers having a larger drainage basin. In terms of length, catchment area and discharge, the Godavari is the largest in peninsular India, and had been dubbed as the Dakshina Ganga (Ganges of the South).

Station

Select Station ▼

Predicted WQI

-

Predicted Grade

-

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Periyar

Periyar (meaning: big river) is the longest river and the river with the largest discharge potential in the Indian state of Kerala. It is one of the few perennial rivers in the region and provides drinking water for several major towns. The Periyar is of utmost significance to the economy of Kerala. It generates a significant proportion of Kerala's electrical power via the Idukki Dam and flows along a region of industrial and commercial activity. The river also provides water for irrigation and domestic use throughout its course besides supporting a rich fishery. Due to these reasons, the river has been named the 'Life-line of Kerala'. Kochi city, in the vicinity of the river mouth draws its water supply from Aluva, an upstream site sufficiently free of seawater intrusion. Twenty five percent of Kerala's industries are along the banks of river Periyar. These are mostly crowded within a stretch of 5 kilometres (3 mi) in the Eloor-Edayar region (Uthiyogamandal), about 10 kilometres (6 mi) north of Kochi harbor.

Station

Select Station ▼

Predicted WQI

-

Predicted Grade

-

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Cauvery

The Kaveri (also known as Cauvery, the anglicised name) is one of the major Indian rivers flowing through the states of Karnataka and Tamil Nadu. The Kaveri River rises at Talakaveri in the Brahmagiri range in the Western Ghats, Kodagu district of the state of Karnataka, at an elevation of 1,341 m above mean sea level and flows for about 800 km before its outfall into the Bay of Bengal. It reaches the sea in Poompuhar, in Mayiladuthurai district. It is the third largest river – after Godavari and Krishna – in southern India, and the largest in the state of Tamil Nadu, which, on its course, bisects the state into north and south. In ancient Tamil literature, the river was also called Ponni (the golden mother, in reference to the fine silt it deposits).

Station

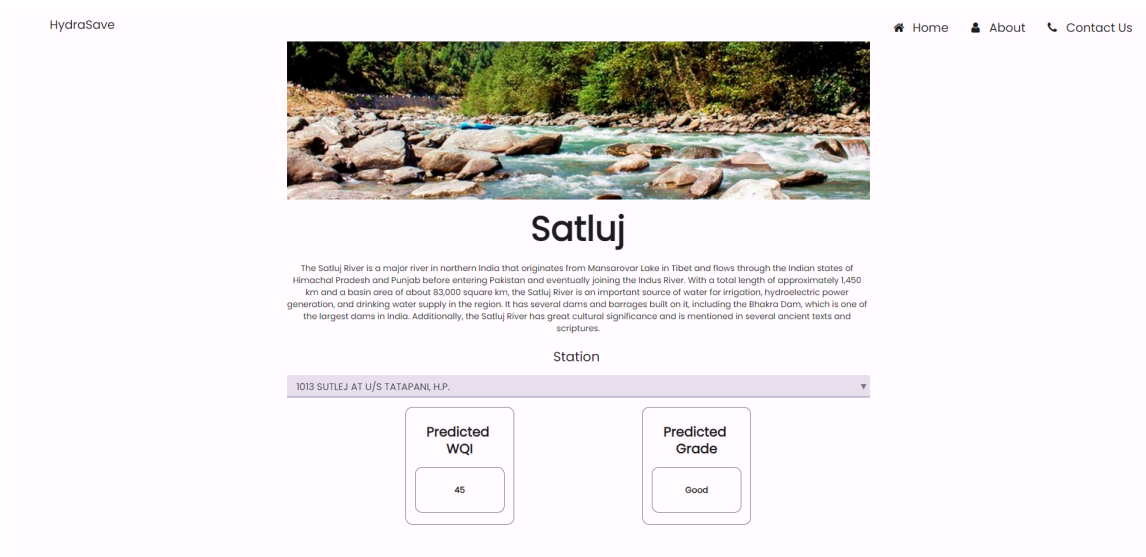
Select Station ▼

Predicted WQI

-

Predicted Grade

-



PUBLICATION

The paper titled “HydraSave : A Unique River Water Quality Forecasting and Pollution Control System”, authored by Ajay Ben Abraham, Aneena Thankachan, Arya Shinod, Devarsh Damodaran and Simy Mary Kurian, was accepted on June 2, 2023, and will be presented at International Conference on Circuit Power and Computing Technologies (IC-CPCT), which will be held at Baselios Mathews II College of Engineering, Sasthamcotta, on August 10 and 11, 2023.



Acceptance Notification- ICCPCT 2023

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Contact:

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On behalf of the Technical Program Committee of the “**International Conference on Circuit Power and Computing Technologies**”, **ICCPCT-2023**, we are pleased to inform you that your paper has been **accepted for oral presentation**. All the papers selected and presented will be published in IEEE Digital Xplore , indexed by Scopus. You can complete the registration process before **5th May 2023**. The conference will be conducted on **10th & 11th August 2023**. The detailed conference schedule will be send to you later. All review comments and suggestions should be incorporated while preparing camera ready paper. Review comments are available in **CMT account>author console>view reviews**. The camera ready paper and copy right should be submitted after **1st May** once IEEE PDF express site gets open. Camera ready paper and copy right form should be submitted via CMT author console and iccpct@gmail.com (mentioning submission number).Follow the instructions given in the IEEE PDF eXpress document for creating PDF eXpress file (available in **website www.iccpct.in in download tab**)

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With regards

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Acknowledgement- Registration completed

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[Quoted text hidden]

HydraSave : A Unique River Water Quality Forecasting and Pollution Control System

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Abstract—Technology has contributed a lot to the betterment of the environment, yet there is still a lot to be done in terms of sustaining nature and its resources. Water pollution is one of the most challenging tasks to be harnessed even with present technology. Thus, continuous monitoring and management of water resources is of prime importance. The river and its tributaries are a vital water source for millions of people. Rivers are not only an important source of water but also considered as habitat for animals and plants. Unfortunately, many rivers and streams are significantly polluted all around the world. Therefore, modelling and predicting water quality is of great importance in controlling water pollution. Water is of different quality and hence it can be used for various purposes. The water quality is measured in terms of WQI (Water Quality Index). With the calculated WQI water can be classified into five different grades. This grading allows us to know for what all purposes water of the predicted quality can be used. Even though there are many existing models for predicting water quality, there is still space for improvements as the existing models focuses only on the learning process and also do not perform missing value imputation. To overcome these problems we are proposing a model which is a combination of CNN (Convolutional Neural Network) and Bi-LSTM (Bidirectional Long Short-Term Memory) models for predicting water quality. The preprocessed data is fed into a WQI Generator whose output WQI value is used for predicting WQI for the next year. This predicted WQI is then fed into the LightGBM (Light Gradient-Boosting Machine) machine for grading the water. The dataset considered for HydraSave is from the water quality data collected from the Kerala State Pollution Control Board (KSPCB). HydraSave can be considered as a supporting hand for the pollution control campaigns conducted by the Government of India. Early prediction allows us to take the necessary steps to prevent water from getting highly polluted. Hence, our proposed model acts as a unique river water quality forecasting and pollution control system.

Index Terms—Bi-LSTM, forecasting, LightGBM, pollution control, WQI

I. INTRODUCTION

In all facets of human, environmental, and societal life, water is a fundamental necessity. The fact that water covers 71% of the world has been thoroughly examined. The rivers are significant natural water supplies. There are many uses for river water, including home and agricultural uses. Water is regarded as the most important natural resource on earth among all other natural resources due to its numerous vital uses. Hazardous wastes are, nevertheless, a result of present urbanisation and industry trends. These wastes, which might take the form of metal, organic pollutants, microbiological bacillus, and other things, are directly dumped into the rivers. Human welfare is directly impacted by water quality. The World Water Assessment Programme states that two million tonnes of human waste are disposed into water courses every day. The global rise in health problems and mortality rates, particularly in developing nations, is attributable to the widespread consumption of contaminated water. According to data from developing countries, over 250 million people contract infections annually, resulting in millions of deaths due to diseases in those regions. According to the National Water Quality Monitoring Program (NWQMP), which is run by the Central Pollution Control Board (CPCB) in collaboration with State Pollution Control Boards (SPCBs) and Pollution Control Committees (PPCs), organic pollution is the primary cause of water pollution at 2500 locations across the nation. According to the extent of organic pollution, the CPCB recognised 150 polluted river sections in 2008, and that number rose to 302 in 2015. The discharge of untreated or only partially treated sewage as well as the discharge of industrial wastewater are the main causes of river pollution.

Hence, it is imperative to maintain regular monitoring of water quality to effectively manage its usage across various sectors. Though various models are available for water quality assessment, these needs further improvement on forecasting

accuracy and speed. Considering the additional requirements for better water quality forecasting we propose a model that combines CNN with Bi-LSTM to provide WQI grading prediction based on sample data followed by grading the water as per its quality using LightGBM technology. The predicted WQI value helps to analyze for what all purposes water of that quality can be used and what measures can be taken to increase the water quality.

II. LITERATURE SURVEY

Various related works were analyzed from the literature being employed for water quality assessment. Conventional methods for water quality monitoring are often burdened with high implementation costs and time-consuming processes for obtaining results, necessitating further improvements in forecasting and validation. In this regard, machine learning (ML) and deep learning (DL) models have garnered attention for their ability to efficiently learn patterns and offer promising performance.

Sakshi Khullar et al. [1] propose a methodology for assessing water quality in a river using Bi-LSTM network. They trained their model using data collected from river Yamuna in India and tested its performance using various evaluation metrics. The results showed that their proposed model achieved high accuracy in predicting water quality, outperforming other traditional models. The model was tested using monthly water quality data collected at various locations in the Delhi region over a period of 6 years (2013-2019). The findings revealed that the model's predicted values were in strong concurrence with the actual values, demonstrating its ability to accurately forecast future trends. Abhishek Kumar et al. [2] proposed another model in their study, where the methodology section detailed the selection of sampling locations, collection of groundwater samples, and analysis of water quality parameters such as pH, electrical conductivity (EC), total dissolved solids (TDS), hardness, alkalinity, chloride, fluoride, nitrate, sulfate, and heavy metals. The authors emphasized the significance of using the WQI as a decision-making tool for groundwater management and policy formulation. The study revealed that around 20% of the area was classified as non-suitable for drinking water, while the remaining areas fell into categories of good, moderate, poor, and very poor based on the WQI classification. The authors also highlighted that the performance of the Artificial Neural Network (ANN) model, although it contains abundant information about the modeled system, is highly dependent on the training strategy employed.

Xiangqian Wang et al. [5] proposed a hybrid model for gas concentration prediction, utilizing the Long Short-Term Memory (LSTM) network and LightGBM algorithm. Their model incorporated a variable weight combination approach to leverage the strengths of both models and improve prediction accuracy. Real-world gas concentration data was used to evaluate the performance of their proposed model, which demonstrated higher accuracy compared to traditional models. This study provided a novel method for gas accident prevention through gas concentration prediction, although further

improvements in performance efficiency may be needed when dealing with rapidly changing input data. On the other hand, HS Atta et al. [9] developed a WQI for assessing the quality of drinking groundwater in the vicinity of the Ismailia Canal in Egypt. They employed a weighted arithmetic mean method to combine the values of various water quality parameters and evaluate the overall groundwater quality. The proposed WQI model was tested on water samples collected from different locations, revealing that the groundwater in the study area was of poor quality and not suitable for drinking purposes. However, it should be noted that Pearson correlation matrix used in the study may not be sufficient to determine nonlinear relationships between variables, and further analysis may be needed to account for such complexities.

SS Baek et al. [10] proposed a novel deep learning model that combined CNN and LSTM networks for the prediction of water level and water quality in rivers. The authors trained their model using data collected from a river in Korea and evaluated its performance using various metrics. The results demonstrated that their proposed model achieved high accuracy in predicting both water level and water quality, surpassing traditional models. However, it should be noted that this study only investigated three pollutants, which is a limited scope compared to existing models. Further research may be needed to expand the scope of pollutants considered in the model.

Md. Galal Uddin et al. [12] presented a comparative study of different WQI models, including their model structures and applications. The study highlighted the parameterization of the models, methods for calculating subindices, parameter weighting values, index aggregation functions, and sources of uncertainty. The WQI model is a popular tool for assessing surface water quality, as it utilizes aggregation techniques to condense large amounts of water quality data into a single value or index. The study also discussed the limitations and issues associated with each model, providing insights into the challenges and considerations in WQI modeling. Yun Ju et al. [6] proposed a combined model of CNN and LightGBM algorithm for ultra-short-term wind power forecasting. The researchers initially analyzed the properties of raw data from time series of the wind field and adjacent wind field to create new feature sets. Subsequently, CNN was used to extract the feature data, and finally, LightGBM classification was employed to enhance the reliability and accuracy of the forecasting. The proposed model was compared with conventional Support Vector Machine (SVM) models, and it demonstrated higher accuracy in wind power forecasting.

From the analysis of existing methods designed for water quality forecasting, it can be concluded that there is still a need for performance enhancement both in terms of classification efficiency and time complexity.

III. PROPOSED METHODOLOGY

This section provides a detailed description of the proposed methodology of HydraSave: A unique river water quality

forecasting and pollution control system. The proposed architecture is shown in fig 1. The system architecture is designed in such a way that it has less computation cost and more efficiency. Initially, the input data is preprocessed and z-

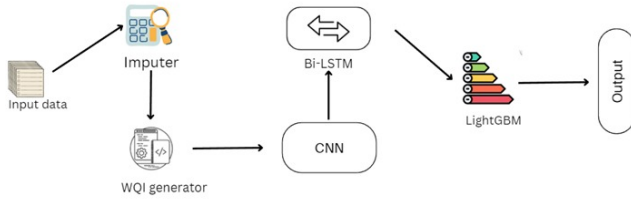


Fig. 1. Proposed architecture of HydraSave

score normalization is performed. This step is done to reduce errors and increase accuracy. This data is then fed into the WQI generator which produces WQI values for the complete dataset. The output values are then fed into CNN and Bi-LSTM models, where prediction of new WQI values are done. The predicted WQI values are classified using the LightGBM classifier which inturn classifies the water into any of the five different grades depending on it's quality. With the output grade achieved, the user will be able to understand for what all purposes water of that quality can be used for the predicted year.

A. Input Data

The input data is collected from the Kerala State Pollution Control Board (KSPCB). The data samples of various rivers of Kerala are analyzed. The dataset used in this study includes various water quality parameters such as Dissolved Oxygen (DO), pH, Biochemical Oxygen Demand (BOD), Total Coliform (TC), and Fecal Coliform (FC), which are indicators of water quality. These parameters are used in the calculation WQI. The data thus obtained is converted into a CSV file for further preprocessing. Forecasting the quality of water is crucial as it enables the implementation of necessary pollution control measures and helps in determining the appropriate usage of water based on its quality grade. The input data collected contains water quality parameters of all stations of the river periyar considered. The time series data of the last 14 years from 2007 to 2020 is analyzed. Detailed information about input data is provided in preliminary analysis.

B. Data Preprocessing

Firstly, the data is passed on to an imputer where missing value imputation is done after which normalization is performed. The steps are described below.

- 1) Imputer : It is important to preprocess the data to avoid errors. This task is performed by the imputer. The missing value imputation technique used in this study involves replacing identified missing values with the mean of neighboring values, using a simple "mean" imputer.

- 2) Z-Score Normalization : A lot of machine learning algorithms attempt to find patterns in data by comparing features of data points. However, there is an issue when the features are on a completely different scales. Therefore, we are performing z-score normalization.

C. WQI Generator

The WQI is a significant tool used to assess the quality of drinking water in urban, rural, and industrial areas. It is calculated based on the composite influence of various water quality parameters that are considered and weighted in the calculation. The "weighted arithmetic index method" is used to compute WQI, where water quality rating (Qi) is an integral part of the index and is determined using the following expression.

$$Q_i = \sum q_i w_i / \sum w_i \text{ --- (1)}$$

Here w_i - Unit weight of i th parameter

q_i - Quality estimate scale of each parameter, it is calculated with the formula -

$$q_i = 100(V_i - V_{Ideal} / S_i - V_{Ideal}) \text{ --- (2)}$$

V_i - Measured value of i th parameter

V_{Ideal} - Ideal value of i th parameter in pure water

S_i - Standard value recommended for i th parameter

w_i is calculated by the formula -

$$w_i = K / S_i \text{ --- (3)}$$

Here K is proportionality constant which is -

$$K = 1 / \sum S_i \text{ --- (4)}$$

Standard value recommended for parameters (S_i)-

DO, mg/L - 10

pH - 8.5

BOD, mg/L - 5

FC/100 mL - 100

TC/100 mL - 1000

Values of unit weights for each parameter (w_i) -

DO pH - 0.2604

BOD - 0.4426

FC - 0.0221

TC - 0.0022

Ideal value of parameters (V_{ideal})

DO pH - 7.0

BOD - 0

FC TC - 0

By using the above method, HydraSave calculates the WQI value for each input data.

D. Convolutional Neural Network

Convolution and pooling operations alternately make up the technique known as CNN. Each neuron executes a dot product on a few inputs, followed by an activation function. It will also have a loss or cost function on the final, completely connected layer, much like in the multilayer perceptron. CNN outperform other neural networks when given inputs such as images, voice, or audio. A standard CNN architecture used for prediction tasks typically comprises multiple layers, such as pooling layers, convolutional layers, and fully connected layers, as shown in Fig 2.

Convolutional layers: These layers apply convolutional filters to the input data to extract features. Each filter is a small matrix of weights that slides over the input data, computing a dot product at each position. The result is a feature map that highlights areas of the input that are important for the prediction task. Convolutional layers typically use non-linear activation functions, such as ReLU, to introduce non-linearity into the model.

Pooling layers: These layers downsample the feature maps produced by the convolutional layers. The most common pooling operation is max pooling, which takes the maximum value within a small window and discards the rest. Pooling helps feature map size reduction and make the model more computationally efficient.

Fully connected layers: These layers take the flattened feature maps from the convolutional and pooling layers and use them to make predictions. Fully connected layers are similar to the layers in a traditional neural network, with each neuron receiving input from neurons of previous layer.

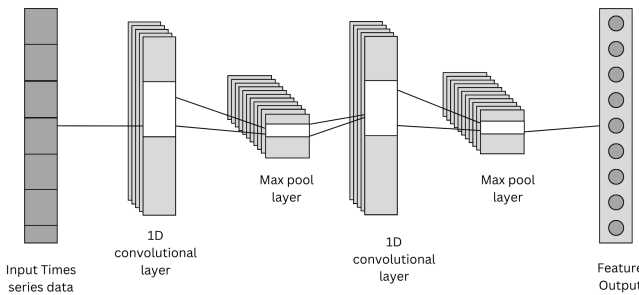


Fig. 2. Structure of CNN

The overall architecture of a CNN for prediction tasks can vary depending on the specific task and dataset. Typically, after the convolutional and pooling layers, one or more fully connected layers are included at the end of the CNN architecture to make predictions. This architecture is often referred to as a "deep" CNN because of its many layers. Overall, CNN architecture for prediction tasks is highly effective because it allows the model to automatically learn and extract features from input data.

E. Bi-LSTM

Bi-LSTM is a variation of the LSTM network. Like a standard LSTM, it can process input sequences and maintain an internal state, but it can also process sequences in both directions, making it bidirectional. The input sequence is fed into the BiLSTM layer, which processes it in two directions. That is, forward and backward. The forward direction processes the input sequence from start to end, while the backward direction processes it from end to start. This allows the BiLSTM to capture context from both the past and the future. At each time step, the BiLSTM layer maintains two internal states which includes a forward state and a backward state. The forward flow is updated based on the previous forward state and the current input, while the backward flow is updated based on the previous backward state and the current input. The output of the BiLSTM layer at each time step is the concatenation of the forward and backward states. The output sequence is then passed to the next layer. By incorporating bidirectional processing, the model is designed to analyze inputs in two directions - one from past to future and the other from future to past. This unique approach allows the LSTM to retain information from both the past and future, as the hidden states are combined. This enables the model to effectively capture contextual information from both directions at any given point in time.

F. CNN with Bi-LSTM

In this study, a combined approach of CNN and Bi-LSTM is employed. While CNN is not well-suited for capturing long-term dependencies in time series data due to its lack of internal state, the Bi-LSTM provides a mechanism to address this limitation. CNN is utilized for extracting spatial features from the input time series data, while the Bi-LSTM is used to capture temporal dependencies. The output of the CNN serves as input to the Bi-LSTM, which generates the forecasted output. The CNN model comprises of two one-dimensional convolutional layers followed by two max-pooling layers, which reduces computational complexity. The 1D convolutional layer generates a feature map using ReLU activation function, followed by max-pooling to downsample and select important features. The resulting output is then combined with the Bi-directional LSTM, which processes the data in both forward and backward directions, facilitating the capture of changes in input parameters.

G. LightGBM

LightGBM is a rapid, distributed gradient boosting framework that employs a tree-based learning algorithm. It is renowned for its support of GPU learning, making it a popular choice for data science application development. One of the distinctive features of LightGBM is its leaf-wise splitting approach as shown in 3, where trees are split based on the best fit at the leaf level, in contrast to other boosting algorithms that split trees depth-wise or level-wise. This vertical tree growth strategy sets LightGBM apart, enabling faster training speed, higher efficiency, lower memory usage, improved accuracy,

and support for parallel, distributed, and GPU learning. Additionally, LightGBM is capable of handling large-scale data, making it a preferred choice for various applications. XGBoost

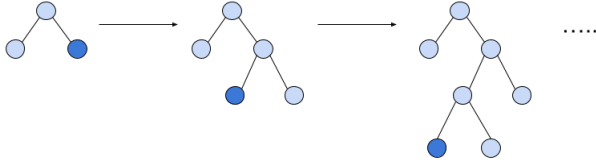


Fig. 3. Leaf-wise growth of LightGBM

utilizes a pre-sorted algorithm and a histogram-based approach for optimal split determination, whereas LightGBM employs a groundbreaking technique known as Gradient-based One-Side Sampling (GOSS) to eliminate data instances (examples or observations). In the histogram-based technique, the split value is determined using discrete bins that group all data points for a particular feature. This approach is faster than the pre-sorted method, which considers all potential split points on the pre-sorted feature values. However, LightGBM's GOSS technique is overall faster than the histogram-based approach.

TABLE I
PREPROCESSED DATA AFTER IMPUTATION

WQI Range	Class	Water Quality
0-25	0	Excellent
25-50	1	Good
50-75	2	Medium
75-100	3	Poor
Above 100	4	Unsuitable for drinking

LightGBM classifies the predicted water quality to any of the above given grades in Table I which indicates for what all purposes water of that quality can be used.

IV. PRELIMINARY ANALYSIS

A. Dataset Collection

The concentration of fecal coliform in Periyar, as recorded by the sewage monitoring station, exceeded the allowable limit of 500mpn/100mL, with an average concentration of 113,000mpn/100mL. Thus, it is important to predict the quality of water which helps to know how much polluted the water is.

The sample data collected contains 5 water quality parameters namely DO, pH, BOD, FC and TC of 8 stations for the last 6 years from 2015 to 2020. The experimental evaluation is conducted by using KSPCB dataset. Several water quality parameters collected from <https://kspcb.kerala.gov.in/water-and-air-quality-directory>.

The sampling stations are

- 1) Eloor
- 2) Kalady
- 3) KWA Aluva



Fig. 4. Map view of Periyar River

- 4) SDP Aluva
- 5) Pathalam
- 6) Kalamassery
- 7) Purappillikkadavu
- 8) Muppathadam

B. Data preprocessing

- 1) Imputer: In the dataset, SDP Aluva with station code 1338 is having missing values which is replaced by the mean of it's neighbourhood values. The values before and after imputation is shown in Table II.

TABLE II
PREPROCESSED DATA AFTER IMPUTATION

Parameter	Before	After
Station	1338	1338
DO	NA	6.4
BOD	NA	1.452
pH	NA	6.622
FC	NA	24615.1
TC	NA	28360.83

- 2) Normalization : For normalization, we used z-score normalization to detect the outliers which are later dropped from the dataset without affecting the performance of the system.

C. Graphical View of The Preprocessed Output Sample

Graphical view of preprocessed output sample is generated to understand and visualize the data after the preprocessed step. The fig 5 Shows the graphical representation of the data.

D. WQI Generation

WQI is computed using the "weighted arithmetic index method". A sample of the generated output is shown in Table III.

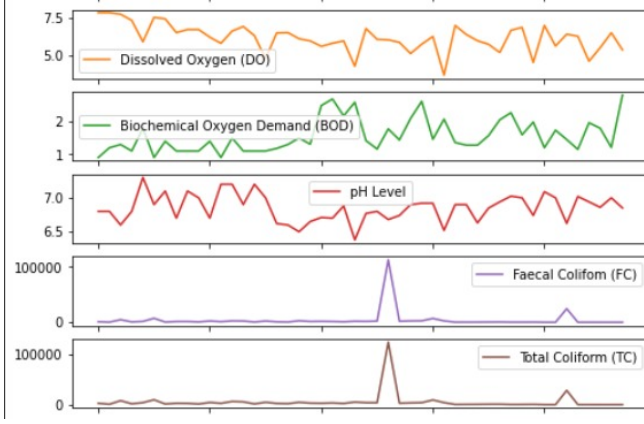


Fig. 5. Graphical View of The Preprocessed Output Sample

TABLE III
SAMPLE DATASET AFTER WQI GENERATION

Station	Location	WQI
18	KALADY	61.187493
3468	KWA ALUVA	47.153553
1338	SDP ALUVA	142.6759
17	ELOOR	99.822425
2333	MUPPATHADAM	57.823608
2334	PATHALAM	202.027134
2336	PURAPPILLIKADAVU	59.639521
2335	KALAMASSERY	75.552203

V. EXPERIMENT AND RESULT

Initially, the generated WQI values are then fed into CNN. The predicted value is again passed to Bi-LSTM for increasing prediction accuracy. The model predicts the WQI value of the year 2020 of all the 8 stations renamed as S1 to S8 respectively. The below provided graph Fig 6 is plotted to visualize the actual and predicted WQI value of each station respectively.

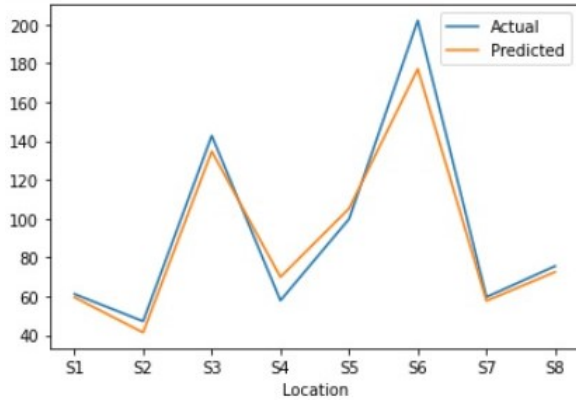


Fig. 6. The graph shows how accurate the result will be for the tested case approximately.

With the predicted value of WQI, the LightGBM classifies the input sample into any of the five different grades. As per the predicted WQI, the LightGBM grades water quality of each

station as shown in Table IV. With the output grade achieved, the user will be able to understand for what all purposes water of that quality is used.

TABLE IV
SAMPLE DATASET AFTER QUALITY GRADING

Station	Location	Grade
18	KALADY	2
3468	KWA ALUVA	1
1338	SDP ALUVA	4
2333	MUPPATHADAM	2
17	ELOOR	4
2334	PATHALAM	4
2336	PURAPPILLIKADAVU	2
2335	KALAMASSERY	2

The performance evaluation is done by various measures like precision, recall, F1-Score. Corresponding values are provided in Table V.

TABLE V
PERFORMANCE MEASURE OF LIGHTGBM

Class	Precision	Recall	F1-Score
0	1.00	1.00	1.00
1	1.00	1.00	1.00
2	1.00	0.91	0.95
3	0.91	1.00	0.95
4	1.00	1.00	1.00

The experimentation result showed that HydraSave requires less computation cost and is much more faster than other existing models. CNN-BiLSTM used in the system showed higher accuracy than CNN-LSTM model. LightGBM classifier has a higher classifying speed when compared to other classification models. Overall, HydraSave is capable of predicting water quality with lesser error rate and higher computation speed.

VI. FUTURE WORK

Though the system shows much better performance when compared, it still can upgrade its working. Some upgradation and changes are expected to be applied in the system for making it highly efficient. It includes:

- 1) Rather than considering river Periyar only, the input data will be extended by taking sample data of other rivers of Kerala.
- 2) Applying transfer learning and thus reduce the overall computation cost.
- 3) Trying to implement using light-weight models, so that the model can be deployed on a mobile device.

VII. CONCLUSION

HydraSave is a water quality forecasting model that utilizes CNN combined with Bi-LSTM networks. The model explores the applications of Bi-LSTM for forecasting WQI by incorporating data pre-processing, normalization, prediction, and classification of water quality. Traditional CNN and LSTM

models often suffer from high computational complexity and low prediction accuracy. To address these limitations, a combination model is used in HydraSave, which helps reduce training errors and improves precision.

Additionally, the HydraSave model incorporates LightGBM technology for water classification based on the predicted WQI value. The model is implemented using time-series data from various monitoring stations of river Periyar in Kerala to study water quality. Accurate prediction of water quality can assist the government in implementing necessary pollution control measures, making HydraSave a unique system for water quality forecasting and pollution control.

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